

SIMATS ENGINEERING ASSESSMENT TOOL 3

Course: MLA03 — Reinforcement Learning for Environment and Agent

Course Outcome: CO3 — Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6) **Annexure:** A — Data Interpretation **Duration:** 1 hr **Questions:** 10 **Total Marks:** 50

Question 1: A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

Interpretation:

The reward trajectory rising steadily from 10 to 45 across episodes is a classic learning curve of a converging RL agent. A monotonic (or near-monotonic) increase shows that the agent is progressively refining its estimate of the value function and, consequently, its policy, rather than acting randomly or oscillating without direction.

Analysis:

In Dynamic-Programming-based and value-iteration-driven RL, each episode allows the agent to update state value or action-value estimates using the Bellman equation, moving these estimates closer to their true (optimal) values. As the value estimates become more accurate, the greedy or near-greedy policy derived from them selects better actions more consistently, which directly raises the cumulative reward obtained per episode. The magnitude of the increase (over 4x, from 10 to 45) suggests the initial policy was far from optimal (heavy exploration / random actions) while the later policy exploits learned knowledge effectively.

Application of Concepts:

This pattern is consistent with the Policy Improvement Theorem, which guarantees that a policy updated greedily with respect to an improved value function is itself no worse, and typically better, than the previous policy. The steady climb evidences repeated cycles of policy evaluation and policy improvement (as in Generalized Policy Iteration), each cycle nudging the agent toward the optimal policy π^* .

Conclusion:

The gradual, sustained rise in reward is strong evidence that the agent is learning effectively: it is converging toward an optimal or near-optimal policy rather than diverging or getting stuck in a poor local behaviour. If the curve begins to plateau near 45, this would further indicate the agent is approaching policy convergence and diminishing returns from further training.

Question 2: An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

Interpretation:

Epsilon (ϵ) controls the probability with which the agent chooses a random action (exploration) instead of the currently best-known action (exploitation). The observed pattern — highest cumulative reward at $\epsilon = 0.1$

and lowest at $\epsilon = 0.5$ — reflects the direct trade-off between exploring unknown actions and exploiting already learned high-value actions.

Analysis:

At $\epsilon = 0.5$, the agent behaves randomly on half of all decisions, so even after the value function has converged reasonably well, it continues to forgo the best-known action 50% of the time, incurring a large opportunity cost and depressing cumulative reward. At $\epsilon = 0.1$, the agent explores only 10% of the time — enough to occasionally discover better actions and avoid getting permanently stuck on a suboptimal choice, while spending 90% of the time exploiting the best-known policy, which maximizes accumulated reward. $\epsilon = 0.3$ sits between these extremes and produces intermediate performance.

Application of Concepts:

This directly applies the exploration–exploitation trade-off central to epsilon-greedy action selection. Excessive exploration (high ϵ) wastes reward-generating opportunities on random, potentially poor actions, while too little exploration (ϵ too close to 0) risks premature convergence to a suboptimal policy because rare, potentially better actions are never sampled. The results show $\epsilon = 0.1$ gives the best balance for this task: sufficient exploration to learn accurate value estimates, combined with high exploitation of the learned policy.

Conclusion:

$\epsilon = 0.1$ provides the best performance among the tested values because it exploits the learned policy most of the time while retaining just enough exploration to keep improving. In practice, this suggests using a low, possibly decaying epsilon (annealing ϵ from a higher value early in training to a low value later) to combine fast early exploration with strong later-stage exploitation.

Question 3: Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

(5 Marks)

Interpretation:

Q-Learning outperforming SARSA by a small but consistent margin across all three checkpoints (100, 200, 300 episodes) reflects the fundamental algorithmic difference between the two: Q-Learning is an off-policy algorithm that updates its value estimates using the maximum reward of the next state's actions, while SARSA is on-policy and updates using the action actually taken by the current (often exploratory) policy.

Analysis:

Because Q-Learning's update rule, $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$, always bootstraps from the best possible next action, it learns the optimal policy's value function directly, even while the agent is still behaving exploratorily (e.g., under ϵ -greedy). SARSA's update, $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot Q(s',a') - Q(s,a)]$, instead reflects the value of the policy actually being followed, including its exploratory (occasionally suboptimal) actions, which tends to produce more conservative, slightly lower reward estimates, especially in environments with risk (e.g., near penalties). This is consistent with the marginally higher rewards obtained by Q-Learning across all episode counts.

Application of Concepts:

This illustrates the off-policy vs. on-policy distinction in temporal-difference control methods. Q-Learning's off-policy, max-based bootstrapping generally converges toward the optimal policy faster and yields higher

achievable reward, whereas SARSA's on-policy learning is safer/more conservative because it accounts for the exploration policy's own risk, often converging more slowly toward the optimal value in reward terms.

Conclusion:

Based on the observed trend, Q-Learning converges faster and achieves consistently higher average reward than SARSA over the same number of episodes. Q-Learning is therefore the more sample-efficient of the two for this task, though SARSA's more cautious, on-policy behaviour can still be preferable in domains where the cost of exploratory mistakes is high.

Question 4: An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

(5 Marks)

Interpretation:

The discount factor γ determines how much weight the agent places on future rewards relative to immediate rewards. The trend of average reward increasing as γ rises from 0.2 to 0.9 shows that this task rewards far sighted, long-horizon planning rather than short-term, myopic decision-making.

Analysis:

With a low γ (0.2), the agent's return, $G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$, is dominated almost entirely by the immediate reward, causing the agent to behave greedily/myopically and ignore delayed but larger rewards, which caps its overall performance. As γ increases toward 0.9, future rewards are discounted much less steeply, so the value function accounts for long sequences of rewards; the agent's policy therefore favours actions that lead to better long-term outcomes even if they sacrifice small immediate rewards, resulting in higher cumulative average reward.

Application of Concepts:

This demonstrates the core role of γ in the Bellman equation and long-term reward optimization: as $\gamma \rightarrow 1$, the agent approximates an undiscounted, far-sighted planner; as $\gamma \rightarrow 0$, it approximates a purely reactive, short-sighted agent. Since environments with delayed rewards (e.g., reaching a distant goal) benefit from valuing future states highly, a higher γ better aligns the learned value function with the true long-term objective.

Conclusion:

$\gamma = 0.9$ is the most suitable discount factor here, as it produces the highest average reward by enabling the agent to optimize for long-term outcomes rather than immediate gains. Values very close to 1 (e.g., 0.99+) could be tested further, but care must be taken since excessively high γ can also slow convergence and increase variance in value estimates.

Question 5: The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

(5 Marks)

Interpretation:

A rise in success rate from 45% at 50 episodes to 92% at 200 episodes represents more than a doubling of task-completion performance and indicates the agent's policy has substantially improved through experience driven value updates.

Analysis:

Early in training (50 episodes), the agent's value function is poorly estimated, so its policy is close to random or weakly informed, explaining the modest 45% success rate. Over the next 150 episodes, repeated policy evaluation and improvement cycles allow the value estimates to converge closer to their true values, so the derived policy makes increasingly correct decisions, driving success rate up to 92%. The steep, sustained rise (not a plateau or oscillation) indicates genuine learning rather than noise.

Application of Concepts:

Applying the Generalized Policy Iteration framework, this progression is what is expected as policy evaluation and improvement steps repeatedly interact: each iteration should produce a policy at least as good as the last. A success rate approaching (but not yet reaching) 100% suggests the agent is close to, but not necessarily exactly at, the optimal policy π^* — some residual randomness (exploration) or incomplete convergence of value estimates likely accounts for the remaining 8% failure rate.

Conclusion:

The agent has clearly learned a substantially improved policy and is trending toward optimality, but 92% success (rather than ~100%) suggests it has not yet fully converged to the optimal policy — further training, reduced exploration, or fine-tuning of learning parameters would likely close this remaining gap.

Question 6: An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.

(5 Marks)

Interpretation:

The learning rate α controls the step size of each value-function update, $Q(s,a) \leftarrow Q(s,a) + \alpha[\text{target} - Q(s,a)]$. The finding that $\alpha = 0.05$ yields the highest stability compared to both 0.01 and 0.1 reflects the classic trade off between learning speed and convergence stability governed by this parameter.

Analysis:

A very low learning rate (0.01) updates value estimates very slowly, so although updates are stable (small, incremental), the agent takes much longer to incorporate new experience, slowing convergence and potentially leaving the CartPole balancing policy under-trained within a fixed number of episodes. A high learning rate (0.1) updates estimates aggressively; each new (possibly noisy) sample can cause large swings in the estimated Q-values, producing oscillation or instability — visible as inconsistent pole-balancing performance in CartPole, where errors compound quickly. $\alpha = 0.05$ strikes a middle ground: updates are large enough to adapt reasonably quickly to new information, but small enough to avoid overshooting and destabilizing the value estimates.

Application of Concepts:

This applies the bias–variance/step-size trade-off inherent to stochastic gradient-style updates in TD learning: too small a step size causes slow, safe but inefficient learning; too large a step size causes fast but noisy, unstable learning. CartPole is especially sensitive to instability because small policy errors can rapidly cascade into pole failure, making a moderate, well-tuned α particularly important.

Conclusion:

$\alpha = 0.05$ provides the best trade-off between learning speed and stability for this CartPole task — fast

enough to converge within a reasonable number of episodes, yet stable enough to avoid the erratic value updates seen at $\alpha = 0.1$. This value is therefore recommended as the most suitable learning rate for reliable convergence.

Question 7: A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario.

(5 Marks)

Interpretation:

The result that ϵ -greedy achieves the highest cumulative reward among ϵ -greedy, softmax, and purely random exploration reflects how well each strategy balances exploring the action space against exploiting already learned high-value actions.

Analysis:

Random exploration selects actions with no regard for their estimated value, so it continues to sample poor actions indefinitely, regardless of how much the agent has already learned — this wastes the largest share of reward-generating opportunities and explains its weak performance. Softmax (Boltzmann) exploration weights action selection by relative estimated Q-values, which is smarter than pure randomness, but it can still assign meaningfully high selection probability to weaker actions when Q-value differences are small, and its performance is sensitive to a temperature parameter that is not always well-tuned. ϵ -greedy, by contrast, is simple and effective: it exploits the single best-known action with probability $(1-\epsilon)$ and explores uniformly only a small fraction of the time (ϵ), giving it a clean, controllable balance that, in this scenario, translated into the highest cumulative reward.

Application of Concepts:

This demonstrates that the effectiveness of an exploration strategy depends on how efficiently it allocates exploratory effort. ϵ -greedy's decisive exploitation of the best current estimate the majority of the time, combined with limited but non-zero exploration, prevented both the reward loss of random exploration and the diluted exploitation seen in softmax when Q-value gaps are narrow.

Conclusion:

ϵ -greedy achieves the best performance in this scenario because it most effectively balances exploration and exploitation for this task's characteristics — reliably exploiting the best-known action while still exploring enough to avoid missing better alternatives. This justifies its selection as the preferred exploration strategy here, though softmax may become more competitive in environments with more nuanced action-value differences requiring finer-grained exploration.

Question 8: An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.

(5 Marks)

Interpretation:

A reduction from 50 steps to 15 steps to reach the goal over 300 episodes is a strong efficiency signal: the agent is learning a shorter, more direct path through the maze, meaning its policy is becoming increasingly goal-directed rather than exploratory or wandering.

Analysis:

Early in training, with little knowledge of the maze's structure, the agent's policy resembles a random walk,

leading to long, inefficient trajectories (50 steps). As episodes progress, repeated visits update the value function so that states leading toward the goal accumulate higher value, and the greedy policy derived from these values increasingly favours the shortest viable path, reducing the step count to 15. The magnitude and consistency of this reduction (over 3x fewer steps) indicates substantial policy improvement rather than incidental variance.

Application of Concepts:

This reflects the Generalized Policy Iteration process converging the value function toward $V^*(s)$, the optimal value function, whose corresponding greedy policy yields the shortest expected path (assuming a step-penalty or shortest-path reward structure). If 15 steps matches (or is very close to) the maze's true shortest possible path, this is strong evidence of convergence to, or very near, the optimal policy π^* .

Conclusion:

The consistent decline to 15 steps strongly suggests the agent has converged to a near-optimal (if not exactly optimal) policy for this maze. To confirm true optimality, this figure should be compared against the maze's known shortest path length; if 15 steps matches that minimum, the agent has effectively solved the task optimally.

Question 9: A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

(5 Marks)

Interpretation:

The SARSA agent achieving its highest long-term reward at $\gamma = 0.95$ (compared to 0.4 and 0.7) shows that this task benefits strongly from valuing delayed rewards almost as highly as immediate ones, since a high discount factor causes the agent's learned value function to weight distant future rewards nearly as much as near-term ones.

Analysis:

At $\gamma = 0.4$, the agent's on-policy return calculation heavily discounts rewards more than a couple of steps ahead, so SARSA's updates, $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma Q(s',a') - Q(s,a)]$, propagate very little information about long-term consequences back to earlier states, causing the agent to favour short-sighted actions and settle for lower long-term reward. At $\gamma = 0.7$, more future information is incorporated, improving performance moderately. At $\gamma = 0.95$, future rewards are barely discounted, so the value function fully reflects the long-

run consequences of actions, allowing SARSA's on-policy learning to identify sequences of actions that maximize cumulative reward over the full episode rather than just the next few steps.

Application of Concepts:

This applies the fundamental role of γ in the return $G_t = \sum \gamma^k r_{t+k+1}$: higher γ shifts the agent's objective toward long-horizon optimization. Because SARSA updates using the value of the action actually taken (on policy), the accurate propagation of long-term reward signal at high γ also helps it learn safer, more consistent long-term behaviour rather than reacting only to immediate incentives.

Conclusion:

$\gamma = 0.95$ is the most effective discount factor here because it enables the agent's policy to be optimized for long-term cumulative reward rather than short-term gains, which best matches tasks where the ultimate payoff depends on a sequence of coordinated actions. Lower discount factors artificially truncate the agent's foresight and therefore underperform.

Question 10: An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

(5 Marks)

Interpretation:

The pattern of highly fluctuating rewards in early episodes followed by stabilization after 500 episodes is characteristic of learning in a stochastic environment, where randomness in transitions and/or rewards initially masks the underlying quality of the agent's policy, but a well-converged policy eventually produces consistently high and stable performance despite this environmental noise.

Analysis:

In early training, the agent's value estimates are inaccurate and still heavily influenced by exploration, so its actions are inconsistent, and combined with inherent environmental stochasticity, this produces large episode to-episode reward variance. As training progresses, repeated sampling allows the agent to average out the effects of randomness and refine its value estimates toward their true expected values (via the law of large numbers implicit in TD/Monte-Carlo updates), so the policy becomes increasingly reliable. Stabilization after 500 episodes indicates the value function has converged sufficiently that the policy consistently selects near optimal actions even when individual transitions are noisy.

Application of Concepts:

This reflects how the Bellman expectation equation implicitly averages over stochastic transitions, so as visit counts to each state-action pair grow, sample-based estimates converge toward the true expected values under the environment's transition dynamics. A stable reward curve post-convergence, despite the environment remaining stochastic, is direct evidence that the policy — not the environment — has become the dominant, reliable factor in determining outcomes.

Conclusion:

The transition from fluctuating to stable rewards demonstrates that the agent has achieved a robust, well converged policy capable of consistently good performance even under environmental uncertainty. This robustness — stability despite stochasticity — is a strong practical indicator of successful policy improvement and learning convergence, rather than the agent being lucky in isolated episodes.

I M Harinath/192425020(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubric for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and	Interprets data correctly with	Partial understanding; misses key trends	Misinterprets or fails to understand data

		relationships	minor gaps		
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand